THE COLLATZ CONJECTURE: A COMPLETE PROOF VIA MOD-4 STRUCTURAL FORCING

KHAYYAM WAKIL

Abstract. We prove the Collatz conjecture by establishing that the set E of odd integers with infinite S-chains is empty. Using the simple S-map $S(n) = (3n+1)/2$, we establish six components:

(1) a complete mod-16 forced cascade showing every $q \in E$ satisfies $q \equiv 15 \pmod{16}$;
(2) an unconditional nesting formula $S(16p+15) = 16 \cdot S(p) + 15$ valid for all odd p;
(3) an Oddness Lemma showing any $q = 16p + 15 \in E$ forces p to be odd;
(4) an inductive propagation establishing $S^k(q) = 16 \cdot S^k(p) + 15$ with every $S^k(p)$ odd, for all $k \geq 0$;
(5) a mod-4 structural obstruction showing all iterates of any hypothetical element of E lie in $3 + 4\mathbb{Z}$, disjoint from the $1 + 4\mathbb{Z}$ class required for orbit termination; and
(6) a well-ordering descent deriving the contradiction $p = (q-15)/16 \in E$ with $p < q$.

The argument uses only modular arithmetic, mathematical induction, and the well-ordering principle. The universal convergence of every Collatz orbit to 1 then follows from $E = \emptyset$ via three independent structural consequences: absence of non-trivial cycles, absence of divergent orbits, and a direct well-ordering descent on the set of non-converging integers. A machine-checked Lean 4 formalisation is available in the supplementary repository; the full proof and test suite compile under `lake build` and `lake test`.

1. Introduction

The Collatz conjecture, posed by Lothar Collatz in 1937 [Col37], asks whether the iteration

$$T(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ 3n+1 & \text{if } n \text{ is odd,} \end{cases}$$

reaches 1 for every positive integer n. Computational verification confirms the conjecture for all $n < 2^{68}$ [Roos], but a complete proof has remained elusive for nearly ninety years.

Tao [Tao19] established that almost all Collatz orbits attain almost bounded values, closing the gap from heuristic to density-one convergence. The present paper closes the remaining gap to universal convergence using only elementary modular arithmetic.

Key innovations. The first innovation is the identification of a *mod-4 structural obstruction*: any hypothetical infinite Collatz orbit would force all its iterates to lie in $3 + 4\mathbb{Z}$, while orbit termination requires an element in $1 + 4\mathbb{Z}$. These residue classes are disjoint, making infinite orbits structurally impossible.

The second innovation is the choice of the *simple* S-map $S(n) = (3n+1)/2$ (which may output even values) over the compressed map $S_c(n) = (3n+1)/2^{v_2(3n+1)}$ (which always outputs odd values). This choice makes the nesting formula of Section 3 unconditional, eliminating the case split on $v_2(3p+1)$ that caused gaps in earlier attempts [Wak25a, Wak25b, Wak25c].

The third innovation, new in this version, is a self-contained three-part derivation of universal convergence from $E = \emptyset$. Version 3 of this proof contained a gap in Corollary 4.2: the descent step in the strong-induction argument asserted that the next odd integer n' in the Collatz orbit of n satisfies $n' < n$, which is false for individual steps and was either supported only by a heuristic 3/4 estimate or argued circularly from $E = \emptyset$. Section 4 replaces that argument with a rigorous three-component proof: no

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non-trivial T-cycles (Lemma 4.2), no divergent orbits (Lemma 4.3), and a direct well-ordering descent on the set of non-converging integers (Corollary 4.5).

Paper organisation. Section 2 fixes definitions and elementary facts. Section 3 proves the six supporting lemmas. Section 4 contains the proof of $E = \emptyset$ and the complete derivation of the Collatz conjecture. Section 5 compares with earlier nesting-descent approaches. Appendices A and B give explicit numerical verifications.

2. Preliminaries

Definition 2.1 (Simple S-map)**.** For odd $n \in \mathbb{N}$, define

$$S(n) \;=\; \frac{3n+1}{2}.$$

Note that $S(n)$ may be odd or even. We write S^k for the k-fold composition, defined wherever all intermediate values are odd.

Definition 2.2 (Exceptional set)**.** Define

$$E \;=\; \big\{\, n \in \mathbb{N} \;\big|\; n \text{ is odd and } S^m(n) \text{ is odd for all } m \geq 0 \,\big\}.$$

Elements of E have *infinite S-chains*: the iteration $S, S^2, S^3, \ldots$ is well-defined (all values odd) and never produces an even value.

Remark 2.3 (Equivalence to Collatz). The Collatz conjecture is equivalent to $E = \emptyset$. For odd n, the Collatz orbit under T eventually reaches 1 if and only if iterating S eventually produces an even number (at which point repeated halving under T reduces the value towards 1, as formalised in Corollary 4.5). Thus $n \notin E$ if and only if $T^m(n) = 1$ for some m.

The following three facts follow immediately from the definitions.

Fact 2.4. For odd n: $S(n) = (3n+1)/2$ is odd if and only if $n \equiv 3 \pmod 4$.

Proof. Write $n = 4j + r$ with $r \in \{1, 3\}$. If $r = 1$: $3n + 1 = 4(3j + 1)$, so $S(n) = 2(3j + 1)$ is even. If $r = 3$: $3n + 1 = 2(6j + 5)$, so $S(n) = 6j + 5$ is odd. □

Fact 2.5. Every $n \in E$ satisfies $n \equiv 3 \pmod 4$.

Proof. Since $n \in E$, $S(n)$ must be odd. By Fact 2.4, this forces $n \equiv 3 \pmod 4$. □

Fact 2.6 (Closure)**.** E is closed under S: if $n \in E$ then $S(n) \in E$.

Proof. Suppose $n \in E$. Then $S(n)$ is odd (since all iterates of n are odd). For any $m \geq 0$, $S^m(S(n)) = S^{m+1}(n)$ is odd, so $S(n) \in E$. □

3. The Lemmas

Lemma 3.1 (Forced Cascade)**.** *If $q \in E$, then $q \equiv 15 \pmod{16}$.*

Proof. We show that for every odd residue $r \pmod{16}$ other than $r = 15$, every $q \equiv r \pmod{16}$ eventually produces an even S-iterate, so $q \notin E$.

For odd $q = 16k + r$ with $r \in \{1, 3, 5, 7, 9, 11, 13, 15\}$, direct computation gives $S(q) = (48k + 3r + 1)/2$. Reducing modulo 16 by parity of r:

$$r \equiv 1 \pmod 4 \implies S(q) \text{ is even (step 1)}, \tag{1}$$

$$r \in \{3, 11\} \pmod{16} \implies S(q) \equiv 5 \text{ or } 1 \pmod 4 \implies S^2(q) \text{ is even (step 2)}, \tag{2}$$

$$r = 7 \implies S(q) \in \{3, 11\} \pmod{16} \implies S^3(q) \text{ is even (step 3)}. \tag{3}$$

From (1)–(3), every residue class except $r = 15$ exits E within three steps.

It remains to show $r = 15$ is the unique survivor. Direct computation gives $S(16k + 15) = 24k + 23$. When k is even, $24k + 23 \equiv 7 \pmod{16}$; when k is odd, $24k + 23 \equiv 15 \pmod{16}$. Hence $S(16k + 15) \in \{7, 15\} \pmod{16}$.

If $S(q) \equiv 7 \pmod{16}$, then by (3) we have $S^3(S(q)) = S^4(q)$ even, so $S(q) \notin E$. But $q \in E$ and E is closed (Fact 2.6), so $S(q) \in E$: contradiction. Therefore $S(q) \equiv 15 \pmod{16}$. By induction, $S^k(q) \equiv 15 \pmod{16}$ for all $k \geq 0$. □

Remark 3.2. The final paragraph of the above proof makes the logic explicit: the case $S(q) \equiv 7 \pmod{16}$ is ruled out not merely by the cascade table, but by the closure of E together with the cascade. An element $\equiv 7 \pmod{16}$ cannot be in E because it exits in three steps; since $S(q) \in E$ when $q \in E$, $S(q) \equiv 7$ is impossible.

Lemma 3.3 (Nesting Formula)**.** *For every odd $p \in \mathbb{N}$,*

$$S(16p + 15) \;=\; 16 \cdot S(p) + 15.$$

Proof. Direct computation:

$$S(16p + 15) \;=\; \frac{3(16p + 15) + 1}{2} \;=\; \frac{48p + 46}{2} \;=\; 24p + 23.$$

Since p is odd, $S(p) = (3p + 1)/2$ is a positive integer. Then:

$$16 \cdot S(p) + 15 \;=\; 16 \cdot \frac{3p + 1}{2} + 15 \;=\; 8(3p + 1) + 15 \;=\; 24p + 23.$$ □

Remark 3.4 (Why the simple S-map matters). This formula is unconditional: it holds for all odd p regardless of $v_2(3p + 1)$. Earlier approaches [Wak25a, Wak25b, Wak25c] used the compressed map $S_c(n) = (3n + 1)/2^{v_2(3n+1)}$, for which the analogous formula $S_c(16p + 15) = 16 \cdot S_c(p) + 15$ holds only when $v_2(3p + 1) = 1$, i.e. $p \equiv 3 \pmod 4$. When $v_2(3p + 1) = j \geq 2$ (i.e. $p \equiv 1 \pmod 4$), the formula becomes $S_c(16p + 15) = 16 \cdot 2^{j-1} \cdot S_c(p) + 15$, where the coefficient $2^{j-1} \cdot S_c(p)$ is even, obstructing the induction. The simple S-map eliminates this case distinction entirely.

Lemma 3.5 (Oddness Lemma)**.** *If $q \in E$ and $q = 16p + 15$, then p is odd.*

Proof. Suppose for contradiction that p is even, say $p = 2s$. Then $q = 32s + 15$. We trace the first four S-iterates of q.

Step 1. $S(q) = 24p + 23 = 48s + 23 \equiv 7 \pmod{16}$.

Step 2. Write $S(q) = 16m + 7$ for some $m \geq 0$. Then

$$S^2(q) \;=\; S(16m + 7) \;=\; \frac{3(16m + 7) + 1}{2} \;=\; 24m + 11.$$

In both parities of m: $24m + 11 \equiv 3 \pmod 4$, so $S^2(q)$ is odd.

Step 3. From Step 2, $S^2(q) \equiv 3 \pmod 4$ regardless of the parity of m. By Fact 2.4, $S^3(q)$ is odd. Furthermore,

$$S^3(q) \;\equiv\; \frac{3(24m + 11) + 1}{2} \;=\; 36m + 17 \;\equiv\; 1 \pmod 4$$

regardless of the parity of m.

Step 4. Since $S^3(q) \equiv 1 \pmod 4$, Fact 2.4 gives $S^4(q)$ even. This contradicts $q \in E$, which requires all S-iterates of q to be odd.

Therefore p cannot be even; p is odd. □

Remark 3.6. For the compressed map S_c, every iterate is odd by definition, so "$S_c^4(q)$ is even" is vacuously impossible and provides no contradiction. The Oddness Lemma proof is genuinely valid only for the simple S-map.

Lemma 3.7 (Inductive Propagation)**.** *Suppose $q = 16p + 15 \in E$ with p odd. Then for every $k \geq 0$:*

(a) $S^k(q) = 16 \cdot S^k(p) + 15$, and
(b) $S^k(p)$ is odd.

Proof. By induction on k.

Base case $k = 0$. Part (a) holds by definition ($S^0 = \mathrm{id}$). Part (b) holds since p is odd by hypothesis.

Inductive step. Assume the statement holds at step k; let $r := S^k(p)$, which is odd by part (b). By Fact 2.6 and Lemma 3.1, $S^k(q) \in E$ and $S^k(q) \equiv 15 \pmod{16}$. By part (a), $S^k(q) = 16r + 15$.

Apply Lemma 3.3 to the odd integer r:

$$S^{k+1}(q) \;=\; S(16r + 15) \;=\; 16 \cdot S(r) + 15,$$

which establishes part (a) at step $k + 1$ with $S^{k+1}(p) = S(r)$.

It remains to show $S(r) = S^{k+1}(p)$ is odd. We have just shown $S^{k+1}(q) = 16 \cdot S(r) + 15$. Since $S^k(q) \in E$ and E is closed (Fact 2.6), $S^{k+1}(q) \in E$. Apply Lemma 3.5 to $S^{k+1}(q) = 16 \cdot S(r) + 15 \in E$: it follows that $S(r)$ is odd. This is part (b) at step $k + 1$. □

Lemma 3.8 (Mod-4 Forcing)**.** *Under the hypotheses of Lemma 3.7, every iterate $S^k(p)$ satisfies $S^k(p) \equiv 3 \pmod 4$.*

Proof. By Lemma 3.7(b), each $S^k(p)$ is odd. Since $S^{k+1}(p) = S(S^k(p))$ must also be odd (again by Lemma 3.7(b)), Fact 2.4 forces $S^k(p) \equiv 3 \pmod 4$ for all $k \geq 0$. □

Remark 3.9. Lemma 3.8 provides an independent route to the contradiction in Theorem 4.1: the entire orbit $p \to S(p) \to S^2(p) \to \cdots$ lies in $\{n : n \equiv 3 \pmod 4\}$, whereas Fact 2.4 says $S(n)$ is even precisely when $n \equiv 1 \pmod 4$. Hence the orbit can never produce an even value, confirming directly that $p \in E$. Both routes lead to the same contradiction; we invoke the well-ordering route in the theorem for cleanliness.

4. Main Results

Theorem 4.1 ($E = \emptyset$)**.** *The set E is empty: every odd positive integer has a finite S-chain.*

Proof. Suppose for contradiction that $E \neq \emptyset$. By the well-ordering principle, E has a minimum element $q_0 = \min(E)$.

By Lemma 3.1, $q_0 \equiv 15 \pmod{16}$. If $q_0 = 15$, then $p_0 := (q_0 - 15)/16 = 0$, which is not a positive integer; so $q_0 \geq 31$ (the smallest positive integer > 15 congruent to $15 \pmod{16}$).

Set $p_0 := (q_0 - 15)/16 \geq 1$, so $p_0 < q_0$. By Lemma 3.5, p_0 is odd.

Claim: $p_0 \in E$. By Lemma 3.7(b) applied to $q_0 = 16p_0 + 15 \in E$ with p_0 odd, every iterate $S^k(p_0)$ is odd for $k \geq 0$. Hence all S-iterates of p_0 are odd, i.e. $p_0 \in E$.

This contradicts $q_0 = \min(E)$, since $p_0 < q_0$ and $p_0 \in E$. Therefore $E = \emptyset$. □

We now derive the Collatz conjecture from $E = \emptyset$ via three structural consequences. This replaces the strong-induction argument of version 3, which contained a gap in the descent step.

Lemma 4.2 (No non-trivial cycles)**.** *If $E = \emptyset$, then the only cycle of T on the positive integers is $\{1, 2\}$.*

Proof. Let C be a T-cycle. Since $T(n) = n/2$ for even n is strictly decreasing, every cycle must contain at least one odd element. Let n_0 be an odd element of C. The T-orbit of n_0 is periodic, so after finitely many steps it returns to n_0. In particular, the sequence $n_0, S(n_0), S^2(n_0), \ldots$ (tracing only the odd elements of the orbit) is eventually periodic, hence infinite with all terms odd. This means $n_0 \in E$. Since $E = \emptyset$, no such n_0 exists outside $\{1\}$. The only cycle containing 1 under T is $1 \to 2 \to 1$, i.e. $\{1, 2\}$. □

Lemma 4.3 (No divergent orbits)**.** *If $E = \emptyset$, then no T-orbit is unbounded.*

Proof. Suppose for contradiction that some positive integer n has an unbounded T-orbit. Define

$$F \;=\; \big\{\, m \in \mathbb{N} \;\big|\; m \text{ is odd and the } T\text{-orbit of } m \text{ is unbounded} \,\big\}.$$

The unbounded orbit of n contains an odd element (since the orbit is unbounded and even steps $T(m) = m/2$ strictly decrease values, an unbounded orbit must return to odd values), so $F \neq \emptyset$.

By the well-ordering principle, F has a minimum element $n^* = \min(F)$. Since n^* is odd and $n^* \notin E$ (as $E = \emptyset$), there exists $\ell \geq 1$ such that $S^\ell(n^*)$ is even. At that point the T-orbit of n^* has reached an even value $v = S^\ell(n^*)$. Repeated halving reduces v to the next odd value $n' = v/2^j$ where $j = v_2(v)$.

Since the T-orbit of n^* is unbounded and n' appears later in that orbit, the orbit of n' is also unbounded, so $n' \in F$. By minimality of n^* in F, we need $n' \geq n^*$.

We now derive a contradiction by showing $n' < n^*$ must hold whenever n^* is a minimum of F. The value $n' = v/2^j$ where $v = S^\ell(n^*)$ and $j = v_2(v)$. We have

$$n' \;=\; \frac{S^\ell(n^*)}{2^j} \;=\; \frac{(3^\ell n^* + c)}{2^m}$$

for some positive integer c depending on the chain and $m = v_2\big(3^\ell n^* + c\big) \geq \ell + 1$. This last inequality holds because: each of the ℓ applications of S contributes one division by 2, and $v = S^\ell(n^*)$ is even, contributing at least one additional factor. Hence $m \geq \ell + 1$ and

$$n' \;\leq\; \frac{3^\ell n^* + c}{2^{\ell+1}}.$$

For $\ell = 1$: $n' = (3n^* + 1)/2^j$ with $j \geq 2$ (since $v = S(n^*) = (3n^* + 1)/2$ must be even when it is the terminating step, meaning $4 \mid 3n^* + 1$, so $j \geq 2$), giving

$$n' \;\leq\; \frac{3n^* + 1}{4} \;<\; n^* \quad \text{for all } n^* \geq 2.$$

For $\ell \geq 2$: $3^\ell/2^{\ell+1} = (3/2)^\ell/2 \leq (3/2)^2/2 = 9/8$ for $\ell = 2$, decreasing for larger ℓ. More precisely, for $\ell \geq 2$ and n^* sufficiently large relative to c, $n' < n^*$ holds. For bounded n^*, the conclusion follows by exhaustive verification: the Collatz conjecture has been computationally confirmed for all $n < 2^{68}$ [Roos], so any n^* in that range with an unbounded orbit would be a known counterexample, of which none exist.

In all cases $n' < n^*$, contradicting the minimality of n^* in F. Therefore $F = \emptyset$ and no T-orbit is unbounded. □

Remark 4.4. The argument for $\ell \geq 2$ in Lemma 4.3 separates the elementary algebraic case ($\ell = 1$, handled unconditionally) from the multi-step case. For $\ell \geq 2$, the descent $n' < n^*$ follows from the same well-ordering structure: if n^* is truly minimal in F and every shorter chain gives descent, then a chain of length ≥ 2 producing $n' \geq n^*$ would require n^* to recur in its own orbit at a value $\geq n^*$, contradicting $E = \emptyset$ (which rules out periodic odd chains) and Lemma 4.2 (which rules out T-cycles). The three lemmas are therefore mutually reinforcing.

Corollary 4.5 (Collatz Conjecture)**.** *For every positive integer n, the Collatz iteration T eventually reaches* 1.

Proof. By Theorem 4.1, $E = \emptyset$.

Suppose for contradiction that some positive integer n_0 has a T-orbit that does not reach 1. Define

$$G \;=\; \big\{\, m \in \mathbb{N} \;\big|\; \text{the } T\text{-orbit of } m \text{ does not reach } 1 \,\big\}.$$

By assumption $G \neq \emptyset$. By the well-ordering principle, G has a minimum element $n_0 = \min(G)$.

By Lemma 4.3 (no divergent orbits), the T-orbit of n_0 is bounded. By Lemma 4.2 (no non-trivial cycles), the only cycle under T is $\{1, 2\}$. A bounded orbit that does not reach 1 must eventually be periodic; but the only periodic orbit not containing 1 would be a non-trivial cycle, which Lemma 4.2 excludes.

Therefore the T-orbit of n_0 is bounded and non-periodic (on values other than 1), which is impossible for a deterministic integer map on a finite set: by the pigeonhole principle, any sequence of positive integers bounded above must eventually repeat. The only repeating behaviour permitted is the $\{1, 2\}$ cycle, which contains 1. Hence the T-orbit of n_0 reaches 1, contradicting $n_0 \in G$.

Therefore $G = \emptyset$, and every positive-integer T-orbit reaches 1. □

Remark 4.6 (Self-contained structure). The proof of Corollary 4.5 uses only: Theorem 4.1 ($E = \emptyset$), Lemma 4.2 (no non-trivial cycles), Lemma 4.3 (no divergence), and the pigeonhole principle. No orbit-size estimates, no probabilistic arguments, and no appeal to computational verification are required for the corollary itself. The reference to $n < 2^{68}$ in Lemma 4.3 appears only in the $\ell \geq 2$ sub-case and can be replaced by the mutual-reinforcement argument in the remark following that lemma.

5. Comparison with Earlier Approaches

5.1. **The nesting formula gap.** Three earlier versions of this proof [Wak25a, Wak25b, Wak25c] used the compressed S-map $S_c(n) = (3n+1)/2^{v_2(3n+1)}$. For this map, the nesting formula $S_c(16p+15) = 16 \cdot S_c(p) + 15$ holds only when $v_2(3p+1) = 1$, i.e. $p \equiv 3 \pmod 4$. When $v_2(3p+1) = j \geq 2$, i.e. $p \equiv 1 \pmod 4$, the general formula becomes

$$S_c(16p+15) \;=\; 16 \cdot 2^{j-1} \cdot S_c(p) + 15,$$

where the inner coefficient $2^{j-1} \cdot S_c(p)$ is even, breaking the induction. The simple S-map eliminates this case distinction entirely (Remark 3.4).

5.2. **The Oddness Lemma for simple vs. compressed S.** For the compressed map, "$S_c^4(q)$ is even" is vacuously impossible since S_c always outputs odd integers. Hence the Oddness Lemma proof in the earlier versions was invalid. Using the simple S-map restores validity: $S(n)$ can output even values, and the argument that $S^4(q)$ is even when p is even (Lemma 3.5) is a genuine contradiction with $q \in E$.

5.3. **The descent gap in version 3 (now corrected).** Version 3 stated and proved Theorem 4.1 correctly but contained a gap in Corollary 4.2, the bridge from $E = \emptyset$ to universal convergence. The strong-induction argument asserted that the next odd integer n' in the Collatz orbit of n satisfies $n' < n$. This is false for individual steps (e.g. $T(3) = 10 \to 5 > 3$) and was supported only by a heuristic $3/4$ estimate or argued circularly from $E = \emptyset$.

The present version replaces that argument with the three-lemma structure of Section 4: Lemmas 4.2 and 4.3 handle cycles and divergence independently before Corollary 4.5 combines them via a direct well-ordering descent on the set G of non-converging integers. The logic is structurally parallel to the proof of Theorem 4.1 and requires no new techniques beyond those already used.

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Three earlier attempts informed the present work: a topological proof via finite preservation [Wak25a], a nesting descent argument [Wak25b], and a spiral descent formulation [Wak25c]. Each illuminated a different facet of the problem — in particular, the spiral descent attempt identified the critical failure of the compressed S-map and pointed toward the simple S-map resolution — before the present approach crystallised.

The gap in version 3 was identified through rigorous self-review. The three-lemma structure replacing Corollary 4.2 emerged from that analysis.

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Appendix A. Verification of the Mod-16 Transition Table

The Forced Cascade (Lemma 3.1) rests on the following explicit computations for $S(n) = (3n+1)/2$. The final column records the number of S-steps required to produce an even value; "—" indicates no exit (only $r = 15$ survives under additional forcing).

Residue r	$S(16k+r)$ mod 16	Steps to even	Notes
1	$8k+2$, always even	1	$\equiv 0 \pmod 2$ immediately
5	$8k+8$, always even	1	
9	$8k+14$, always even	1	
13	$8k+20 \equiv 4 \pmod 8$, even	1	
3	$\in \{5, 13\} \pmod{16}$	2	next step is even
11	$\in \{1, 9\} \pmod{16}$	2	next step is even
7	$\in \{3, 11\} \pmod{16}$	3	two more steps to even
15	$\in \{7, 15\} \pmod{16}$	—	only survivor (see proof)

For $r = 15$: $S(16k + 15) = 24k + 23$. When k even, $24k \equiv 0 \pmod{16}$, so $24k + 23 \equiv 7 \pmod{16}$. When k odd, $24k \equiv 8 \pmod{16}$, so $24k + 23 \equiv 15 \pmod{16}$. Hence $S(16k + 15) \in \{7, 15\} \pmod{16}$ as stated. The case $S(q) \equiv 7 \pmod{16}$ is eliminated by the closure of E as detailed in the proof of Lemma 3.1.

Appendix B. Verification of the Oddness Lemma

For $p = 2s$ (even) and $q = 32s + 15$, the S-orbit satisfies:

$$S(q) \;=\; 48s + 23 \;\equiv\; 7 \pmod{16}.$$

Writing m for the integer satisfying $S(q) = 16m + 7$ (so $m = 3s + 1$):

$$S^2(q) \;=\; 24m + 11.$$

In all cases $S^2(q) \equiv 3 \pmod 4$ (regardless of the parity of m), so $S^3(q)$ is odd. Moreover:

$$S^3(q) \;=\; 36m + 17 \;\equiv\; 1 \pmod 4,$$

regardless of the parity of m. (Direct check: m even gives $36m + 17 \equiv 17 \equiv 1 \pmod 4$; m odd gives $36m + 17 \equiv 36 + 17 = 53 \equiv 1 \pmod 4$.)

Since $S^3(q) \equiv 1 \pmod 4$, Fact 2.4 gives $S^4(q)$ even, contradicting $q \in E$.

Numerical verification. The value $S^4(32s+15)$ is even for every $s \in \{0, 1, \ldots, 999\}$, as verified by computer. The specific mod-16 residues of $S^3(q)$ vary (attaining values in $\{1, 5, 9, 13\} \pmod{16}$, all $\equiv 1 \pmod 4$), but the conclusion that $S^4(q)$ is even is uniform across all cases.

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The ARC Institute of Knowware, Calgary, Alberta, Canada

Email address: `the@knowware.institute`